PRACTICAL 6 — CAPSTONE SOC WORKFLOW SIMULATION

1. Objective

The objective of this practical is to simulate a complete SOC incident lifecycle, including attack simulation, detection, triage, response, escalation, and reporting. This demonstrates end-to-end Security Operations Center (SOC) workflow handling.

2. Tools Used

- Wazuh (simulated SIEM for detection)
- Metasploit (attack simulation concept only)
- TheHive (incident tracking simulation)
- Google Docs / Excel (reporting and documentation)

3. Attack Simulation

A simulated attack was performed targeting a vulnerable Samba service in a test environment (Metasploitable2 scenario).

Attack Details

Field	Value
Attack Type	Remote Code Execution
Target System	Metasploitable2 (Simulated)
Source IP	192.168.1.101
Vulnerability	Samba usermap script
MITRE Technique	T1210 (Exploitation of Remote Services)

4. Detection & Triage

The simulated SIEM environment detected suspicious activity targeting Samba services.

Detection Table

Timestamp	Source IP	Alert Description	MITRE Technique
2026-01-13 14:00:00	192.168.1.101	Samba exploit attempt	T1210

5. Response & Containment

Immediate incident response actions were performed:

- The affected system was isolated from the network
- The attacker's IP was blocked at the firewall level
- Logs and evidence were preserved for forensic investigation
- Incident escalated to Tier 2 SOC team

6. Escalation Report (TheHive Simulation)

- **Incident Case Details**

Field	Value
Case ID	INC-101
Severity	Critical
Status	Escalated to Tier 2
Analyst	Tier 1 SOC

A critical security incident was detected involving a suspected Samba service exploitation attempt from IP address 192.168.1.101. The activity aligns with MITRE ATT&CK technique T1210, indicating exploitation of remote services. Initial detection was triggered by abnormal network behavior targeting vulnerable SMB services. The affected system was immediately isolated to prevent further compromise. The malicious IP was blocked at the network perimeter, and relevant logs were preserved for forensic analysis. Due to the severity and potential impact, the incident has been escalated to Tier 2 for in-depth investigation and root cause analysis.

7. Reporting

Executive Summary

A simulated attack targeting a vulnerable Samba service was successfully detected and mitigated using SOC procedures. The incident was contained before any damage occurred.

Timeline

- 14:00 — Attack detected
- 14:01 — SIEM alert triggered
- 14:02 — System isolated
- 14:05 — IP blocked
- 14:10 — Escalation to Tier 2

Recommendations

- Disable unused SMB services
- Apply regular security patches
- Monitor remote service access logs
- Implement detection rules for MITRE T1210 patterns

8. Manager Briefing

A security attempt was detected where an attacker tried to exploit a vulnerable service on a system. The attack was quickly identified and blocked before causing any damage. The system was isolated, and the attacker's access was prevented. The incident has been escalated for further investigation. No data loss or service disruption occurred, and normal operations remain unaffected.

9. SOC Relevance

This practical demonstrates:

- End-to-end SOC incident lifecycle
- Attack simulation and detection mapping
- Incident response and containment
- Escalation workflow handling
- Reporting for technical and non-technical stakeholders

10. Conclusion

The simulated attack scenario was successfully handled through detection, containment, and escalation procedures. This demonstrates a complete SOC workflow aligned with industry-standard incident response practices and MITRE ATT&CK framework mapping.